This is a long title

Adelaide Diamictite*[1], Petra Sandstone[2], Dover Limestone[3], Jasper[1,2], and Cowell Jade[1]

[1]The Other Test Institute of Sedimentologika, Adelaide, Australia
[2]The Test Institute of Sedimentologika, Bern, Switzerland
[3]The Third Test Institute of Sedimentologika, Geneva, Switzerland

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*Corresponding author: Adelaide Diamictite, thisisnotarealemail@notvalid.org

Abstract

This is a test abstract used for evaluating the build of an article using the *Sedimentologika* LaTeX template using LuaLaTeX. It contains no meaningful scientific content, but contains useful guidelines for writing your article using the template. **Your** abstract will need to fit within the first page. You can use \documentclass[preprint]{sedimentologika} to test if your abstract is too long. If too long, they will print over the text in the lower left of the first page.
We encourage authors to provide a translation of your abstract into a language of your choice. If you wish to provide multiple translations, please contact the editorial team.
If your translation language requires special fonts (e.g., Arabic), instructions are available in the template file and the readme.md file. Should you need further help with this, please contact the editorial team admin@sedimentologika.org.

This preprint was generated using the *Sedimentologika* LaTeX template

Plain language summary

This file tests the *Sedimentologika* article template build using LuaLaTeX. A plain language summary is mandatory.

Résumé

Il s'agit d'un résumé fictif utilisé pour évaluer la compilation d'un article à l'aide du modèle LaTeXde *Sedimentologika*avec LuaLaTeX. Il ne contient aucun contenu scientifique significatif, mais fournit des instructions utiles pour rédiger votre article à l'aide de ce modèle.

1 Introduction

As of 2026, *Sedimentologika* has moved to LaTeX to typeset accepted publications in the journal. The primary motivations for this switch are to:

- use an open-source, free software for typesetting,
- provide authors with a streamlined pre-print to publication workflow,
- lower the time investment required by our volunteer typesetting team, and,
- ensure all papers published in *Sedimentologika* have a unified design.

While the primary uses of the template are for typesetting by the *Sedimentologika* team and authors preparing their article for submission to *Sedimentologika* it also provides you with an easy way to generate a pre-print version that you can post on a pre-print server (e.g., EarthArXiv). Simply use \documentclass[preprint]{sedimentologika} in the preamble above.

Like the policy of Seismica, we do not require authors to use this template at initial submission. **However**, minimum formatting requirements apply and upon acceptance authors using LaTeX should submit their final version using the *Sedimentologika* LaTeX template. Not using the template will incur delays in copy-editing, proofing, and publication, For authors using Microsoft Word or LibreOffice etc, there will be greater delays incurred depending on the complexity of the document.

2 Minimum formatting and layout requirement

While *Sedimentologika* strongly encourages the use of our templates for articles submitted to the journal, you may submit your article without using the templates if the following conditions are met:

- Your article is written in English
- Your manuscript is a single PDF, DOC, DOCX, or ODT file containing all figures and in-text tables, in their appropriate place.
- Your data is made accessible via a repository such as Zenodo.
- You have added to your manuscript the following statements, after the conclusions and before the references
- Data and code availability
- Conflict of interest statement
- Artificial intelligence use statement
- Funding

3 Language

Sedimentologika encourages the use of

4 Figures and tables

This section outlines the general guidelines for figures and tables. Please refer to *Sedimentologika*'s figure and table guidelines for further detail.

The final document will have a two-column page layout. You can test the appearance of your tables and figures in this layout by using the `preprint` option for the `sedimentologika` document class.

4.1 Dimensions

It is best practice to design and proof your figures and tables at their intended final size. Figures and tables should preferably be designed to fit within either a singe column (e.g., Figure 1) or across two columns (e.g., Figure 2) in portrait page mode. However, landscape figures and tables are allowable, but only use these if absolutely required for legibility. Table 1 summarises the maximum dimensions allowable for figures and tables that fit within a single page.

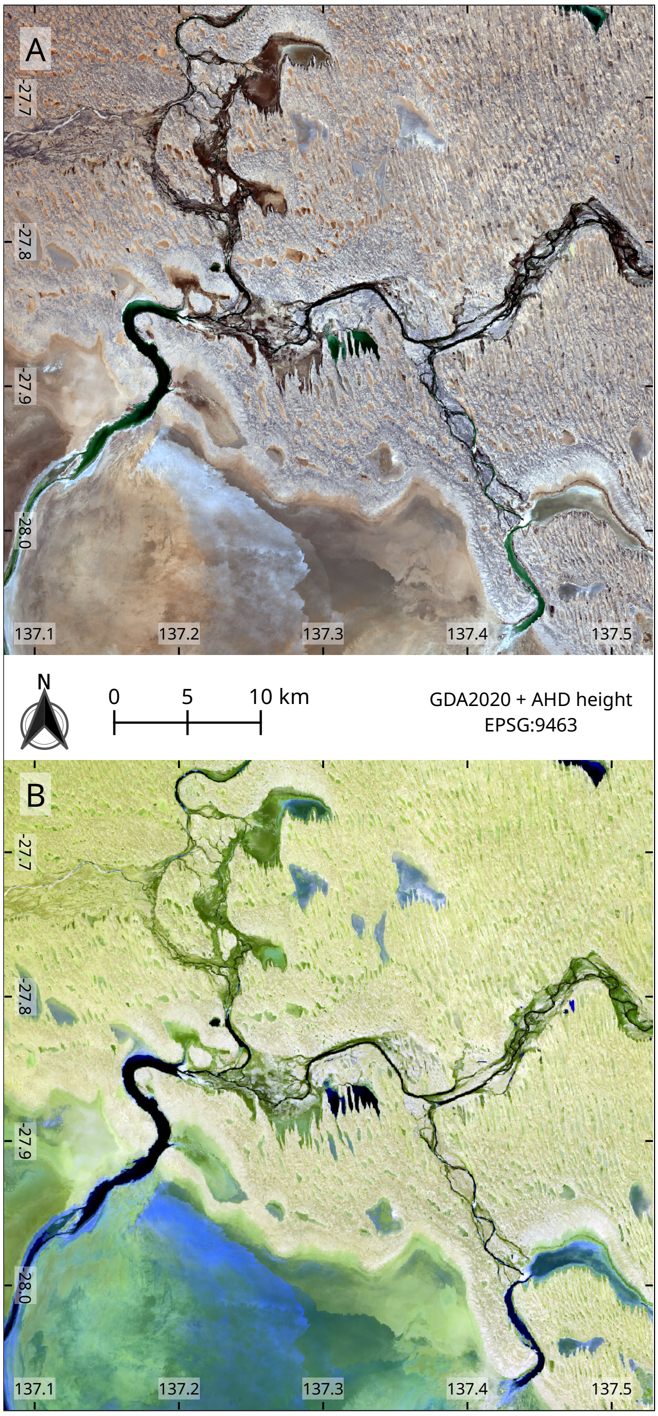

Figure 1 – Sentinel 2 L2A imagery (2025-10-18) of north-eastern Kati Thanda-Lake Eyre at the Warburton River discharge. **(A)** True colour image mapping bands 4, 3, and 2 to R, G, and B channels. **(B)** False colour image mapping bands 12, 11, and 2 to R, G, and B channels. Contains modified Copernicus Sentinel data [2025].

Intended Width	Maximum Width	Maximum Height
one column	86 mm	230 mm
two column	180 mm	230 mm
landscape	257 mm	160 mm

Table 1 – Summary of standard dimensions for figures and tables.

If a figure is being placed at the end of the document rather than near its insertion point use `\clearpage` directly after the figure environment.

4.2 Figure design

Figures must be designed with accessibility in mind. Do not rely on colour alone, use symbols or patterns to assist with differentiation on graphs and other diagrams.

Preferably use the Noto Sans font family, however any standard, sans-serif font (e.g., Arial, Helvetica, Verdana) can be used. Be consistent and only use one font family.

Use at most three (preferably no more than two) different font sizes for annotations on figures. The standard font size should be 12 pt, and the absolute minimum font size should be 6 pt depending on the font family, or 2 mm at final production size. Aim to use larger font sizes (e.g., 12 pt) to improve accessibility.

Colour schemes must be colour-vision deficiency safe.

4.3 Maps

Map figures must have a scale, grid markers of some form, and state the coordinate reference system of their projection. Figure 1 is an exemplar of minimum requirements for map figures.

Due to the nature of geology being an international science, some maps may have areas of disputed borders or regions. In these cases, please use the internationally recognised border or region where possible.

4.4 Indexing and cross-referencing

Figures and tables should be numbered in order of appearance and cross-referenced in text. In LaTeX, use `\autoref{ref}` to handle this. For example, these are cross-references to Figure 2 and Table 1 using `autoref`.

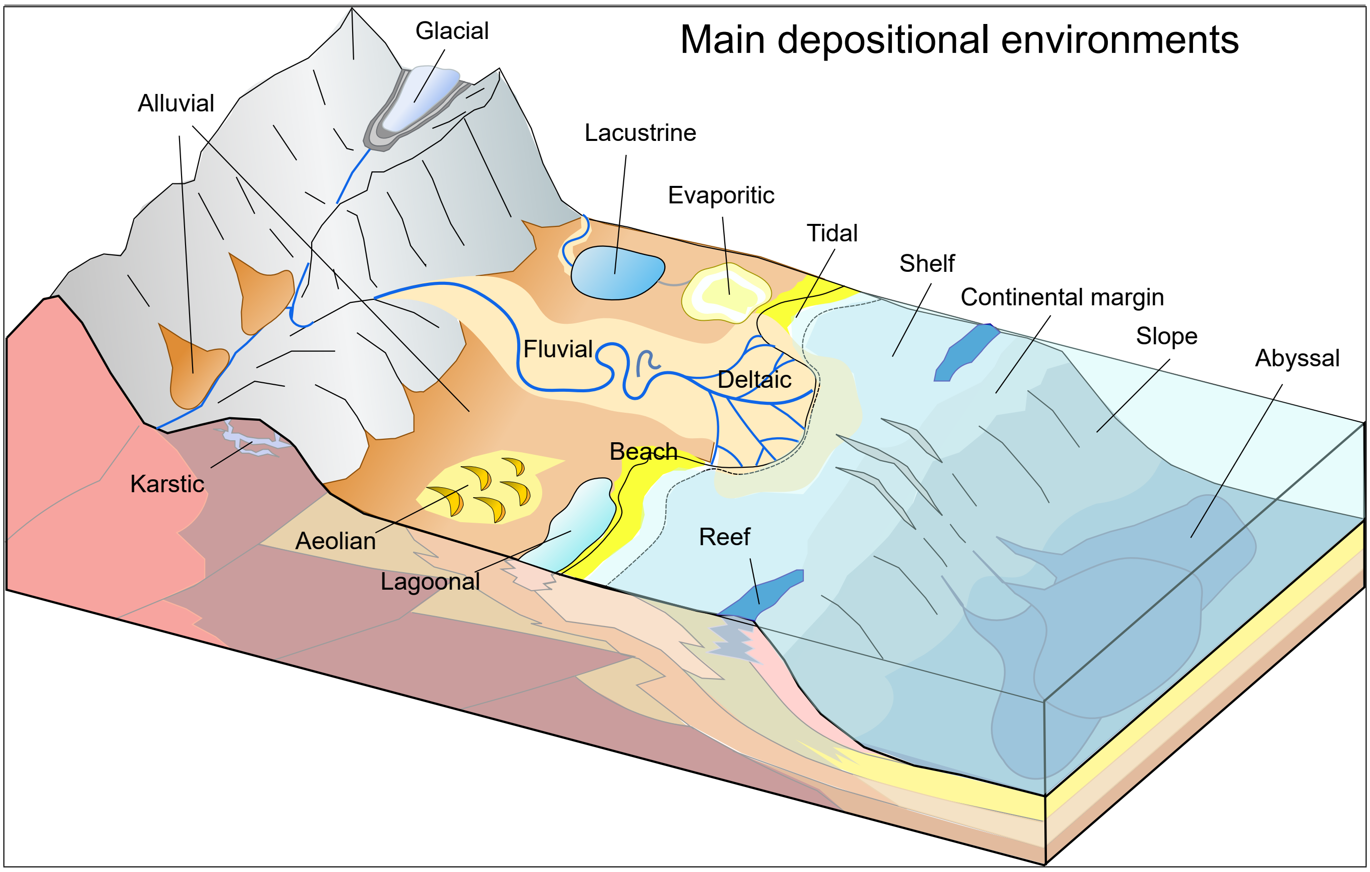

Figure 2 – Schematic diagram of the main depositional environments in sedimentary systems. Source: Mikenorton (2018)

For figures with sub-panels, add uppercase letters for indexing purposes as shown in Figure 1A and Figure 1B. If additional sub-indices are required, use parenthesised lowercase Roman numerals, e.g., A(i).

4.5 Table formatting

Tables rules are formatted using booktabs. This means you can use the `\toprule`, `\midrule`, and `\bottomrule` commands within the `tabular` environment.

The `NiceTabular*` package is also loaded, providing any easy ability to have a column automatically adjust it's spacing to be flexible. Use `X` in the width of the column(s) you want to be flexibly adjusted.

Table 1 is a simple example of a one-column wide table with correct styling and using the `NiceTabular*` environment.

To ensure correct widths of tables, please use the provided custom commands

- `\onecolwidth` for single-column width
- `\twocolwidth` for two-column width

as the size of your `NiceTabular*` or `tabular*` environment.

For tables that will extend beyond a single page, `longtable` is available. If the `longtable` needs to be two columns wide, or in landscape orientation, please see the `example_longtable.tex` and `example_landscape_table.tex` for an example of how to format these.

To place a table that is only a single page column wide (i.e. small tables), use the `table` environment.

Otherwise, use the `table*` environment to place a table that will span both page columns. For all `table*` environments, precede them with a `\clearpage` command.

5 Mathematics

In general follow the Short Math Guide (PDF).

Use `\vec{x}` for vectors and `\mathbf{M}` for matrices, e.g.

```
\vec{y} = \mathbf{X}\vec{\beta} + \epsilon
```

will print:

$$\vec{y} = \mathbf{X}\vec{\beta} + \epsilon \tag{1}$$

In-line equations will look like this: $x^2 + y^2 = z^2$, while basic display equations will look like the following:

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{i=n} x_j = \frac{x_1 + x_2 \ldots + x_n}{n} \tag{2}$$

Multiline equations will look as follows:

$$\begin{multline}\hat{\rho}^2_{OP}\left(\rho^2\right) = 1 - \frac{n-3}{n-(k+1)-1}\cdot \\ \left(1-\rho^2\right){}_2F_1\left(1,1;\frac{n-(k+1)-1}{2};1-\rho^2\right) \tag{3}\end{multline}$$

Finally, aligned equations will look like this:

$$\begin{align}\sum\left(x_i-\varepsilon_1\right)\left(x_i-\varepsilon_2\right)\left(x_i-\varepsilon_3\right)\left(x_i-\varepsilon_4\right) &= 0 \tag{4}\\ \sum x_i\left(x_i-\varepsilon_1\right)\left(x_i-\varepsilon_2\right)\left(x_i-\varepsilon_3\right)\left(x_i-\varepsilon_4\right) &= 0 \tag{5}\\ \sum {x_i}^2\left(x_i-\varepsilon_1\right)\left(x_i-\varepsilon_2\right)\left(x_i-\varepsilon_3\right)\left(x_i-\varepsilon_4\right) &= 0 \tag{6}\\ \sum {x_i}^3\left(x_i-\varepsilon_1\right)\left(x_i-\varepsilon_2\right)\left(x_i-\varepsilon_3\right)\left(x_i-\varepsilon_4\right) &= 0 \tag{7}\end{align}$$

6 Referencing

Appropriate citation of work is mandatory. While peer-reviewed literature should be the primary sources on which you rely, material considered to be *grey literature* such as pre-prints, theses, government reports, conference papers etc. should be appropriately cited.

The *Sedimentologika* class uses the citation-style-langauge package for referencing. Use `\cite{}` for parenthetical citations, e.g., (Strata, 2065), and `\textcite{}` for in-line text citations, e.g., Helios et al. (2072) stated that this is an awesome template. Prefixes, suffices, and pointers can be added as optional arguments, e.g., `\cite[prefix = {e.g., }, suffix={and references therein}]{ExampleThesis}` will produce (e.g., Diamond, 2060 and references therein).

The following is to enable generation of a reference list with examples for an article (Helios et al., 2072), book (Strata, 2065), book section (Core, 2068), dataset (Cryo & Ice, 2075), map (Carto, 2069), preprint (Plume & Geyser, 2074), report (Agency for Outer Worlds Geology, 2070), software (OrbitLab, 2073) and thesis (Diamond, 2060).

7 Geologic Time

For geologic time, please use nomenclature of the International Chronostratigraphic Chart, the latest version can be found at text.

8 Conclusions

This is a mandatory section, replace this text.

Thanks for using the *Sedimentologika* LaTeX template for preparing your article.

Acknowledgements

Replace this text with your acknowledgements.

Data and code availability

Data used in this paper is hosted on Zenodo at doi.org/10.1000/182, `\cite{ExampleCitation}`.

The code used in this paper is available via and archived on Zenodo at doi.org/10.1000/182, `\cite{ExampleCitation}`.

Supplementary materials

If you have any additional supplementary material hosted in a repository for this manuscript please replace this with a statement linking to the doi, and add the citation, e.g. Supplementary material for this paper can be found at doi.org/10.1000/182 `\cite{ExampleCitation}`.

Conflict of interest statement

The authors declare they do not have any tangible or perceived conflicts of interest with regard to this research.

Artificial intelligence use statement

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